

Lecture #6: Continuous Probability Distributions

MATH 145 – Calculus I

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The Normal Distribution

A random variable Y is said to have a **normal (or Gaussian) probability distribution** if and only if, for $\sigma > 0$ and $-\infty < \mu < \infty$, the density function of Y is

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}, \quad -\infty < y < \infty.$$

We write

$$Y \sim \text{Normal}(\mu, \sigma^2)$$

to say that Y is drawn from a normal distribution with parameters μ and σ^2 .

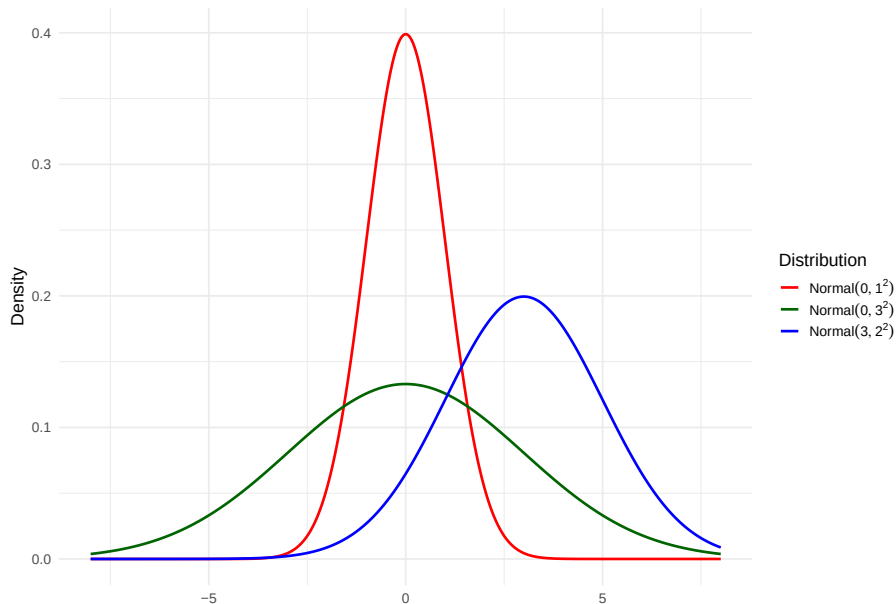
The Support of a Random Variable

The **support** of a random variable is the range of values the random variable can take on with non-zero probability.

In the case of the binomial distribution, the support of Y is $\{0, 1, 2, \dots, n\}$.

In the case of the normal distribution, the support of Y is $\mathbb{R}$.

The Graph of the Normal Distribution



Facts about Definite Integrals

A function f is **even** if $f(x) = f(-x)$ for all $x \in \mathbb{R}$.

If $f(x)$ is **even**, then for any $a > 0$:

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

A function f is **odd** if $-f(x) = f(-x)$ for all $x \in \mathbb{R}$.

If $f(x)$ is **odd**, then for any $a > 0$:

$$\int_{-a}^a f(x) dx = 0$$

Expected Value and Variance of the Normal Distribution

Theorem: If $Y \sim \text{Normal}(\mu, \sigma^2)$, then

$$E[Y] = \mu \qquad \text{and} \qquad V[Y] = \sigma^2$$

Proof of Expected Value of the Normal Distribution

$$\begin{aligned} E[Y] &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} y \exp \left\{ -\frac{(y-\mu)^2}{2\sigma^2} \right\} dy \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (u + \mu) \exp \left\{ -\frac{u^2}{2\sigma^2} \right\} du \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} u \exp \left\{ -\frac{u^2}{2\sigma^2} \right\} du \\ &\quad + \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \mu \exp \left\{ -\frac{u^2}{2\sigma^2} \right\} du \\ &= 0 + \mu \left(\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \exp \left\{ -\frac{u^2}{2\sigma^2} \right\} du \right) \\ &= \mu. \end{aligned}$$

Proof of Variance of the Normal Distribution

$$\begin{aligned} V[Y] &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (y - \mu)^2 \exp\left\{-\frac{(y - \mu)^2}{2\sigma^2}\right\} dy \\ &= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-\frac{z^2}{2}} dz \\ &= \frac{\sigma^2}{\sqrt{2\pi}} \left(-ze^{-\frac{z^2}{2}} \Big|_{-\infty}^{\infty} \right) + \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz \\ &= 0 + \sigma^2 \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz \right) \\ &= \sigma^2 \end{aligned}$$

The Standard Normal Distribution

The **standard Normal (or Gaussian) random variable** Z has a PDF

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{z^2}{2} \right\}.$$

That is, $Z \sim \text{Normal}(0, 1)$ is a Normal random variable with mean $\mu = 0$ and $\sigma^2 = 1$.

The cumulative distribution function (CDF) of the standard Normal is defined as the function $\Phi(\cdot)$

$$\Phi(z) \stackrel{\text{def}}{=} F_Z(z) = P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{t^2}{2}} dt.$$

CDF of an Arbitrary Normal Distribution

Theorem (CDF of an arbitrary Normal): Let $X \sim \text{Normal}(\mu, \sigma^2)$.
Then

$$F_X(x) = P(X \leq x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

Examples:

- 1 Find $P(Z \leq 2.00)$.

CDF of an Arbitrary Normal Distribution

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Then

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Examples:

- ① Find $P(Z \leq 2.00)$.
- ② Let $X \sim \text{Normal}(3, 4)$. Find $P(X \leq 11)$.

Properties of Normal Distributions

Theorem: Let $X \sim \text{Normal}(\mu, \sigma^2)$. Then the following results hold:

- ① $\Phi(y) = 1 - \Phi(-y)$.
- ② $P(X \geq b) = 1 - \Phi\left(\frac{b-\mu}{\sigma}\right)$.
- ③ $P(|X| \geq b) = 1 - \Phi\left(\frac{b-\mu}{\sigma}\right) + \Phi\left(\frac{-b-\mu}{\sigma}\right)$.

Skewness and Kurtosis

For a random variable X with PDF $F_X(x)$, define the following central moments as:

$$\text{mean: } E[X] \stackrel{def}{=} \mu,$$

$$\text{variance: } E[(X - \mu)^2] \stackrel{def}{=} \sigma^2,$$

$$\text{skewness: } E\left[\left(\frac{X - \mu}{\sigma}\right)^3\right] \stackrel{def}{=} \gamma,$$

$$\text{kurtosis: } E\left[\left(\frac{X - \mu}{\sigma}\right)^4\right] \stackrel{def}{=} \kappa.$$

Skewness and Kurtosis

The **kurtosis** of a distribution deals with the “peakedness” or “tailedness” of the distribution.

We define **excess kurtosis** to be $\kappa - 3$. Excess kurtosis is easier to understand/interpret than “raw” kurtosis.

The normal distribution has skewness 0 and excess kurtosis 0.

Percentiles of Probability Distributions

Let X be a continuous random variable with probability density function f_X . The **median of X** is a point $c \in \mathbb{R}$ such that

$$F_X(c) = \int_{-\infty}^c f_X(x) dx = \int_c^{\infty} f_X(x) dx.$$

Theorem: The **median** of a random variable X is the point c such that

$$F_X(c) = \frac{1}{2}.$$

Percentiles of Probability Distributions

What about other percentiles, like the 35th percentile?

The k^{th} **percentile** of a random variable X is the point c such that

$$F_X(c) = \frac{k}{100}.$$

The **quantile function** is the inverse of the CDF. These are hard to compute for most distributions, so we tend to use R.